

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems COURSE

CODE: CSA0509

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COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems.* (BL3, BL6)

Activity Tool : Debugging & Optimisation Task (Low) Assessment Tool

Weightage: 10%

QUESTIONS		
Q.N o	Question	Bloom's Level
1	Identify and correct errors in a given SQL CREATE TABLE statement with incorrect syntax and missing constraints.	BL2 – Understand
2	Debug an SQL query that incorrectly uses aggregate functions without GROUP BY and fix the issue. Bloom's Level:	BL2 – Understand
3	Correct a faulty SELECT query where column names are misspelled or improperly referenced.	BL1 – Remember
4	Fix errors in an SQL query using JOIN, where the join condition is missing or incorrect.	BL3 – Apply

5	Identify and correct a subquery error where multiple values are returned instead of a single value.	BL2 – Understand
6	Debug a VIEW creation statement that contains invalid column references or syntax errors.	BL1 – Remember
7	Examine an SQL query with unnecessary conditions and optimize it by simplifying the WHERE clause.	BL3 – Apply
8	Identify violations of integrity constraints in a given table and correct them.	BL2 – Understand
9	Fix a relation that violates 1NF or 2NF by removing repeating groups or partial dependencies.	BL3 – Apply
10	Identify redundant attributes in a relation and apply basic normalization (up to 3NF) to improve design.	BL3 – Apply

QUESTION:1 Identify and correct errors in a given SQL CREATE TABLE statement with incorrect syntax and missing constraints.

ANSWERS:

```
mysql> CREATE DATABASE DebugDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE DebugDB;
Database changed
```

ERROR:

```
mysql> CREATE TABLE Student_Info student_id INT, student_name VAR
CHAR, marks, PRIMARY KEY student_id;
ERROR 1064 (42000): You have an error in your SQL syntax; check t
he manual that corresponds to your MySQL server version for the r
ight syntax to use near 'student_id INT, student_name VARCHAR, ma
rks, PRIMARY KEY student_id' at line 1
```

CORRECTED:

```
mysql> CREATE TABLE Student_Info (student_id INT PRIMARY KEY, student_name VARCHAR(100), marks DECIMAL(5,2));
Query OK, 0 rows affected (0.04 sec)
```

QUESTION: 2 Debug an SQL query that incorrectly uses aggregate functions without GROUP BY and fix the issue. Bloom's Level:

ANSWERS:

```
mysql> CREATE TABLE Staff (staff_id INT PRIMARY KEY, staff_name VARCHAR(100), dept_name VARCHAR(50), salary DECIMAL(10,2));
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Staff VALUES (1,'John','HR',40000),(2,'Alice','HR',45000),(3,'Bob','IT',60000),(4,'David','IT',65000),(5,'Eva','Sales',50000);
Query OK, 5 rows affected (0.03 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

ERROR:

```
mysql> SELECT dept_name, AVG(salary) FROM Staff;
ERROR 1140 (42000): In aggregated query without GROUP BY, expression #1 of SELECT list contains nonaggregated column 'debugdb.Staff.dept_name'; this is incompatible with sql_mode=only_full_group_by
```

CORRECTED:

```
mysql> SELECT dept_name, AVG(salary) FROM Staff GROUP BY dept_name;
+-----+-----+
| dept_name | AVG(salary) |
+-----+-----+
| HR        | 42500.000000 |
| IT        | 62500.000000 |
| Sales     | 50000.000000 |
+-----+-----+
3 rows in set (0.01 sec)
```

QUESTION: 3 Correct a faulty SELECT query where column names are misspelled or improperly referenced.

ANSWERS:

```
mysql> CREATE TABLE Employee (emp_id INT PRIMARY KEY, emp_name VARCHAR(100), department VARCHAR(50), salary DECIMAL(10,2));
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Employee VALUES (1,'Rahul','HR',35000),(2,'Priya','IT',50000),(3,'Arun','Sales',42000);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

ERROR:

```
mysql> SELECT emp_nam, departmnt, salry FROM Employee;
ERROR 1054 (42S22): Unknown column 'emp_nam' in 'field list'
```

CORRECTED:

```
mysql> SELECT emp_name, department, salary FROM Employee;
+-----+-----+-----+
| emp_name | department | salary |
+-----+-----+-----+
| Rahul    | HR         | 35000.00 |
| Priya    | IT         | 50000.00 |
| Arun     | Sales      | 42000.00 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

QUESTION: 4 Fix errors in an SQL query using JOIN, where the join condition is missing or incorrect.

ANSWERS:

```
mysql> CREATE TABLE Department (dept_id INT PRIMARY KEY, dept_name VARCHAR(50));
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Department VALUES (1,'HR'),(2,'IT'),(3,'Sales');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT*FROM Department;
+-----+-----+
| dept_id | dept_name |
+-----+-----+
| 1       | HR        |
| 2       | IT        |
| 3       | Sales     |
+-----+-----+
3 rows in set (0.00 sec)

mysql> ALTER TABLE Staff ADD dept_id INT;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE Staff SET dept_id=1 WHERE staff_id=1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Staff SET dept_id=1 WHERE staff_id=2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Staff SET dept_id=2 WHERE staff_id=3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

ERROR:

```
mysql> SELECT s.staff_name,d.dept_name FROM Staff s JOIN Departme
nt d ON s.department_id=d.dept_id;
ERROR 1054 (42S22): Unknown column 's.department_id' in 'on claus
e'
```

CORRECTED:

```
mysql> SELECT s.staff_name,d.dept_name FROM Staff s JOIN Departme
nt d ON s.dept_id=d.dept_id;
+-----+-----+
| staff_name | dept_name |
+-----+-----+
| John      | HR        |
| Alice     | HR        |
| Bob       | IT        |
| David     | IT        |
| Eva       | Sales     |
+-----+-----+
5 rows in set (0.00 sec)
```

QUESTION: 5 Identify and correct a subquery error where multiple values are returned instead of a single value.

ANSWERS:

```
mysql> CREATE TABLE Departments (dept_id INT PRIMARY KEY, dept_name VARCHAR(50));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> INSERT INTO Departments VALUES (1,'HR'),(2,'IT'),(3,'Sales');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> CREATE TABLE Staffs (staff_id INT PRIMARY KEY, staff_name VARCHAR(100), dept_name VARCHAR(50), salary DECIMAL(10,2), dept_id INT);
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Staff VALUES (1,'John','HR',40000,1),(2,'Alice','HR',45000,1),(3,'Bob','IT',60000,2),(4,'David','IT',65000,2),(5,'Eva','Sales',50000,3);
ERROR 1062 (23000): Duplicate entry '1' for key 'staff.PRIMARY'
mysql> INSERT INTO Staffs VALUES (1,'John','HR',40000,1),(2,'Alice','HR',45000,1),(3,'Bob','IT',60000,2),(4,'David','IT',65000,2),(5,'Eva','Sales',50000,3);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Staffs;
```

staff_id	staff_name	dept_name	salary	dept_id
1	John	HR	40000.00	1
2	Alice	HR	45000.00	1
3	Bob	IT	60000.00	2
4	David	IT	65000.00	2
5	Eva	Sales	50000.00	3

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Departments;
```

dept_id	dept_name
1	HR
2	IT
3	Sales

```
3 rows in set (0.00 sec)
```

ERROR:

```
mysql> SELECT staff_name FROM Staff WHERE dept_id=(SELECT dept_id FROM Department);
ERROR 1242 (21000): Subquery returns more than 1 row
```

CORRECTED:

```
mysql> SELECT staff_name FROM Staff WHERE dept_id IN (SELECT dept_id FROM Department);
+-----+
| staff_name |
+-----+
| John      |
| Alice     |
| Bob       |
| David     |
| Eva       |
+-----+
5 rows in set (0.00 sec)
```

QUESTION: 6 Debug a VIEW creation statement that contains invalid column references or syntax errors.

ANSWERS:**ERROR:**

```
mysql> CREATE VIEW Staff_View AS SELECT s.staff_name,d.department
FROM Staff s JOIN Department d ON s.dept_id=d.dept_id;
ERROR 1054 (42S22): Unknown column 'd.department' in 'field list'
```

CORRECTED:

```
mysql> CREATE VIEW Staff_View AS SELECT s.staff_name,d.dept_name
FROM Staff s JOIN Department d ON s.dept_id=d.dept_id;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM Staff_View;
+-----+-----+
| staff_name | dept_name |
+-----+-----+
| John      | HR        |
| Alice     | HR        |
| Bob       | IT        |
| David     | IT        |
| Eva       | Sales     |
+-----+-----+
5 rows in set (0.00 sec)
```

QUESTION: 7 Examine an SQL query with unnecessary conditions and optimize it by simplifying the WHERE clause.

ANSWERS:**ORIGINAL (UNOPTIMIZED):**

```
mysql> SELECT * FROM Staff WHERE salary>50000 AND salary>=30000 AND dept_name='IT' AND dept_name='IT';
```

staff_id	staff_name	dept_name	salary	dept_id
3	Bob	IT	60000.00	2
4	David	IT	65000.00	2

```
2 rows in set (0.00 sec)
```

CORRECTED (OPTIMIZED):

```
mysql> SELECT * FROM Staff WHERE salary>50000 AND dept_name='IT';
```

staff_id	staff_name	dept_name	salary	dept_id
3	Bob	IT	60000.00	2
4	David	IT	65000.00	2

```
2 rows in set (0.00 sec)
```

QUESTION: 8 Identify violations of integrity constraints in a given table and correct them.

ANSWERS:

```
mysql> INSERT INTO Orders VALUES (101,NULL,'Completed');
ERROR 1146 (42S02): Table 'debugdb.orders' doesn't exist
mysql> CREATE TABLE Orders (order_id INT PRIMARY KEY, customer_id INT NOT NULL, status VARCHAR(20) CHECK(status IN('Pending','Shipped','Delivered')));
Query OK, 0 rows affected (0.04 sec)
```

ERROR:

```
mysql> INSERT INTO Orders VALUES (101,NULL,'Completed');
ERROR 1048 (23000): Column 'customer_id' cannot be null
```

CORRECTED:

```
mysql> INSERT INTO Orders VALUES (101,1,'Pending');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT*FROM Orders;
```

order_id	customer_id	status
101	1	Pending

```
1 row in set (0.00 sec)
```

QUESTION: 9 Fix a relation that violates 1NF or 2NF by removing repeating groups or partial dependencies

ANSWERS:

ERROR:


```
mysql> CREATE TABLE Employee_Project (emp_id INT, emp_name VARCHAR(100), project1 VARCHAR(50), project2 VARCHAR(50), project3 VARCHAR(50));
ERROR 1050 (42S01): Table 'employee_project' already exists
mysql> CREATE TABLE Employee_Project (emp_id INT, emp_name VARCHAR(100), project1 VARCHAR(50), project1 VARCHAR(50));
ERROR 1050 (42S01): Table 'employee_project' already exists
```

CORRECTED:

```
mysql> CREATE TABLE Employee_Projects (emp_id INT, emp_name VARCHAR(100), project1 VARCHAR(50), project2 VARCHAR(50), project3 VARCHAR(50));
Query OK, 0 rows affected (0.04 sec)
```

QUESTION: 10 Identify redundant attributes in a relation and apply basic normalization (up to 3NF) to improve design.

ANSWERS:

```
mysql> CREATE TABLE Order_Details (order_id INT, customer_name VARCHAR(100), customer_city VARCHAR(50), product_name VARCHAR(100), product_price DECIMAL(10,2), quantity INT);
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Customer (customer_id INT PRIMARY KEY, customer_name VARCHAR(100), customer_city VARCHAR(50));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Customer (customer_id INT PRIMARY KEY, customer_name VARCHAR(100), customer_city VARCHAR(50));
ERROR 1050 (42S01): Table 'customer' already exists
mysql> CREATE TABLE Product (product_id INT PRIMARY KEY, product_name VARCHAR(100), product_price DECIMAL(10,2));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Orders (order_id INT PRIMARY KEY, customer_id INT, FOREIGN KEY(customer_id) REFERENCES Customer(customer_id));
ERROR 1050 (42S01): Table 'orders' already exists
mysql> CREATE TABLE Order_Item (order_id INT, product_id INT, quantity INT, PRIMARY KEY(order_id,product_id), FOREIGN KEY(order_id) REFERENCES Orders(order_id), FOREIGN KEY(product_id) REFERENCES Product(product_id));
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> INSERT INTO Customer VALUES (1,'Rahul','Chennai'),(2,'Priya','Coimbatore');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Customer;
+-----+-----+-----+
| customer_id | customer_name | customer_city |
+-----+-----+-----+
|          1 | Rahul         | Chennai       |
|          2 | Priya         | Coimbatore    |
+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> INSERT INTO Product VALUES (101,'Laptop',55000.00),(102,'Mouse',800.00),(103,'Keyboard',1500.00);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Product;
+-----+-----+-----+
| product_id | product_name | product_price |
+-----+-----+-----+
|         101 | Laptop       | 55000.00      |
|         102 | Mouse        | 800.00         |
|         103 | Keyboard     | 1500.00        |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

Code of Conduct:

I M. Vishnu Vardhan Reddy (192411053))_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M. Vishnu Vardhan Reddy